

Top Quark Signal in the Cosmological Higgs Collider

1 Setup and conventions

We compute the connected Higgs four-point function generated by the top Yukawa box in the cosmological Higgs-collider setup [4], in exact de Sitter space with constant m_t and zero chemical potential. We use the free two-component construction of Aoki et al. [1], Eqs. (2.5)–(2.29), but derive the four-vertex contractions and their collapsed expansion separately. Their bubble correlator, and in particular its Yukawa bispectrum cancellation, is not a box amplitude. The organization follows the gauge-boson note: setup, lines and vertices, box contraction, soft expansion, scalar integrals, and signal.

1.1 Geometry, indices and the Majorana decomposition

Our metric is $\eta_{ab} = (+, -, -, -)$ and

$$ds^2 = a^2(d\tau^2 - d\mathbf{x}^2), \quad a = -\frac{1}{H\tau}, \quad \sigma^a = (\mathbf{1}_2, \boldsymbol{\sigma}), \quad \bar{\sigma}^a = (\mathbf{1}_2, -\boldsymbol{\sigma}), \quad \tau < 0. \quad (1)$$

For undotted and dotted indices alike take

$$E = (\epsilon_{\alpha\beta}) = i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (\epsilon^{\alpha\beta}) = -E, \quad \psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta, \quad \psi_\alpha = \epsilon_{\alpha\beta}\psi^\beta. \quad (2)$$

In particular $\psi\chi = \psi^\alpha\chi_\alpha = \psi^T E\chi = \chi\psi$ for Grassmann fields, and, for the column $d = (\psi_1^\dagger, \psi_2^\dagger)^T$, $(\psi\psi)^\dagger = -d^T E d$. A dagger reverses operator order. We Fourier transform *every field*, including d , with $e^{i\mathbf{k}\cdot\mathbf{x}}$ and measure $d^3k/(2\pi)^3$; thus two-field contractions carry $\delta^3(\mathbf{k} + \mathbf{k}')$. This differs from taking the adjoint of a Fourier coefficient, which reverses its momentum. A prime on a correlator removes the overall $(2\pi)^3\delta^3(\sum \mathbf{k}_i)$.

Let t_{cov} be the left-handed top, and t_{cov}^c the left-handed charge conjugate of the right-handed top: $t_{\text{cov}}^c = [(t_R)^C]_L$, with two-component form $i\sigma^2 t_R^*$ up to the Yukawa phase convention. It carries the conjugate gauge representation; it is not a second right-handed field. For $\Phi = (0, (h_0 + \delta h)/\sqrt{2})^T$, define

$$g_t = y_t/\sqrt{2}, \quad m_t = g_t h_0, \quad \mu = m_t/H, \quad N_c = 3. \quad (3)$$

The spin connection disappears on setting $t = a^{3/2}t_{\text{cov}}$ and $t^c = a^{3/2}t_{\text{cov}}^c$. The fermion density, with the coordinate measure $d\tau d^3x$, becomes

$$\mathcal{L} = it^\dagger \bar{\sigma}^\mu \partial_\mu t + it^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(m_t + g_t \delta h)(tt^c + \text{h.c.}). \quad (4)$$

The remaining a is $a^4(a^{-3/2})^2$, not a^3 or a^4 . Use Aoki's change of variables

$$t = \frac{\psi_1 + i\psi_2}{\sqrt{2}}, \quad t^c = \frac{\psi_1 - i\psi_2}{\sqrt{2}}. \quad (5)$$

The cross terms cancel because $\psi_1\psi_2 = \psi_2\psi_1$; the kinetic cross terms cancel between the two original fields. Consequently

$$\mathcal{L} = \sum_{I=1}^2 \left[i\psi_I^\dagger \bar{\sigma}^\mu \partial_\mu \psi_I - \frac{a}{2}(m_t + g_t \delta h)(\psi_I \psi_I + \text{h.c.}) \right]. \quad (6)$$

The two copies are identical and independent. This decomposition applies color by color after gauge interactions are omitted. It is not a splitting into two separately gauge-covariant colored Majorana particles: gauge transformations mix the two real components of the original charged Dirac representation. A connected loop belongs to one copy throughout, so we calculate one copy and restore $2N_c$ once.

quantity	this note	Aoki et al.
metric	$(+, -, -, -)$	$(-, +, +, +)$
Clifford convention	$\{\gamma^a, \gamma^b\} = 2\eta^{ab}$	$\{\gamma^a, \gamma^b\} = -2\eta^{ab}$
rescaled Weyl field	ψ_I	$\tilde{\psi}_I$
mass index	$\mu = m_t/H$	m with $H = 1$
mode variable	$x = -k\tau$	$2ik\tau = -2ix$
propagators	rescaled fields	covariant fields in (2.23)–(2.29)
SK contractions	no prefactor i	no prefactor i

The Pauli matrices, epsilon convention, and Fourier sign agree. The numerical gamma matrices can be the same despite the opposite metric signatures. To convert our two-point functions to Aoki's covariant ones multiply by $(a_1 a_2)^{-3/2} = H^3(\tau_1 \tau_2)^{3/2}$; do not also change the rescaled vertex.

1.2 External Higgs fields

The existing light spectator-Higgs prescription is retained:

$$G_s(k; \tau) = \frac{H^2}{2k^3} (1 - isk\tau) e^{isk\tau}, \quad s = \pm 1. \quad (7)$$

We require $|\dot{h}_0| \ll H|h_0|$ and $|m_{h,\text{eff}}^2| \ll H^2$ over the relevant times. A nonzero constant h_0 does not itself guarantee a light fluctuation; its renormalized potential and curvature mass are part of this assumption. We compute $\langle \delta h^4 \rangle'_c$ at order g_t^4 , without specifying a Higgs-to-curvature conversion. A linear Yukawa interaction has no elementary $\delta h^2 \psi \psi$ vertex.

2 Two-component modes, propagators and vertices

2.1 Helicity modes

For one copy, suppress I and color. Use Aoki's normalized spinors

$$h_+(\hat{k}) = \begin{pmatrix} \cos(\theta/2) \\ e^{i\phi} \sin(\theta/2) \end{pmatrix}, \quad h_-(\hat{k}) = \begin{pmatrix} -e^{-i\phi} \sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}, \quad P_\lambda = h_\lambda h_\lambda^\dagger = \frac{\mathbf{1}_2 + \lambda \boldsymbol{\sigma} \cdot \hat{k}}{2}. \quad (8)$$

The operator expansion, with $\{b_\lambda(\mathbf{k}), b_{\lambda'}^\dagger(\mathbf{k}')\} = (2\pi)^3 \delta_{\lambda\lambda'} \delta^3(\mathbf{k} - \mathbf{k}')$, is

$$\psi_\alpha(\tau, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \sum_{\lambda=\pm} \left[h_{\lambda\alpha} u_\lambda b_\lambda(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}} + (E h_\lambda^*)_\alpha v_\lambda^* b_\lambda^\dagger(\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (9)$$

There is one set of $b, b^\dagger$ per Majorana copy. Both u_λ and v_λ below are positive-frequency *radial* functions. In particular v_λ is not the negative-frequency four-component mode called v in the old note. The field equation yields

$$i\partial_\tau u_\lambda + \lambda k u_\lambda = a m_t v_\lambda, \quad i\partial_\tau v_\lambda - \lambda k v_\lambda = a m_t u_\lambda, \quad (10)$$

$$u_\lambda'' - a H u_\lambda' + (k^2 + a^2 m_t^2 + i\lambda a H k) u_\lambda = 0, \quad v_\lambda'' - a H v_\lambda' + (k^2 + a^2 m_t^2 - i\lambda a H k) v_\lambda = 0. \quad (11)$$

With $x = -k\tau$, the normalized BD solutions are

$$\boxed{u_+ = v_- = \frac{\mu}{\sqrt{2x}} W_{-1/2, i\mu}(-2ix), \quad u_- = v_+ = \frac{i}{\sqrt{2x}} W_{1/2, i\mu}(-2ix).} \quad (12)$$

Take the boundary value from the BD continuation. At large x the nonvanishing massless-helicity amplitudes tend to $e^{i\pi/4} e^{ix}$, and the other amplitudes are $O(\mu/x)$. The first-order equations conserve $|u_\lambda|^2 + |v_\lambda|^2$, which is one by this boundary condition. Together with $EP_\lambda^T(-\hat{k})E^{-1} = P_\lambda(\hat{k})$ this gives the canonical anticommutators, including $\{\psi, \psi\} = 0$.

It is useful to introduce only two auxiliary radial functions

$$F = \frac{\sqrt{\pi x}}{2} e^{\pi\mu/2} H_{1/2-i\mu}^{(1)}(x), \quad G = \frac{\sqrt{\pi x}}{2} e^{-\pi\mu/2} H_{1/2+i\mu}^{(1)}(x). \quad (13)$$

Aoki's Whittaker–Hankel identities (2.19) give the exact dictionary

$$u_\lambda = e^{3i\pi/4} \frac{F - \lambda G}{\sqrt{2}}, \quad v_\lambda = e^{3i\pi/4} \frac{F + \lambda G}{\sqrt{2}}, \quad -iF' = \frac{\mu}{x} F + G, \quad -iG' = F - \frac{\mu}{x} G. \quad (14)$$

The primes in the last two relations mean d/dx . This is a change of radial variables, not the introduction of an additional fermion species. The fixed common phase cancels from all contractions. The $\mu = 0$ modes are defined by continuity.

2.2 All lower-index contractions and SK components

Let $\zeta_+ = \psi$ and $\zeta_- = d$ denote *vertex types*, not SK branches, and write $C_{\alpha\beta}^{rs,>}(\mathbf{k}; \tau_1, \tau_2) = \langle \zeta_{r\alpha}(\tau_1, \mathbf{k}) \zeta_{s\beta}(\tau_2, -\mathbf{k}) \rangle'$. An index on a type— slot is dotted; numerical matrix notation identifies the two bases using $\sigma^0 = \mathbf{1}_2$. For $u_j = u_\lambda(k, \tau_j)$ and $v_j = v_\lambda(k, \tau_j)$, all four 2×2 contractions follow directly from the one nonzero vacuum product $\langle b b^\dagger \rangle$:

$$\begin{aligned} C^{++,>} &= - \sum_\lambda u_1 v_2^* P_\lambda E, & C^{+-,>} &= \sum_\lambda u_1 u_2^* P_\lambda, \\ C^{-+,>} &= - \sum_\lambda v_1 v_2^* E P_\lambda E, & C^{--,>} &= \sum_\lambda v_1 u_2^* E P_\lambda. \end{aligned} \quad (15)$$

For example $\langle \psi_\alpha \psi_\beta \rangle' = \sum_\lambda u_1 v_2^* h_{\lambda\alpha} \epsilon_{\beta\gamma} h_{\lambda\gamma}^*$; the minus sign in $-P_\lambda E$ is $E^T = -E$. These anomalous contractions do not vanish for one Majorana copy. Their cancellation in $\langle tt \rangle$ follows instead from the relative i^2 between the two independent copies in Eq. (5).

The signed lesser contraction and the four SK components are

$$\begin{aligned} C^{rs,<}(\mathbf{k}; 1, 2) &= -[C^{sr,>}(-\mathbf{k}; 2, 1)]^T, \\ C_{-+}^{rs} &= C^{rs,>}, & C_{+-}^{rs} &= C^{rs,<}, \\ C_{++}^{rs} &= \theta(\tau_1 - \tau_2)C^{rs,>} + \theta(\tau_2 - \tau_1)C^{rs,<}, & C_{--}^{rs} &= \theta(\tau_2 - \tau_1)C^{rs,>} + \theta(\tau_1 - \tau_2)C^{rs,<}. \end{aligned} \quad (16)$$

$$(17)$$

Here the two subscripts are always SK labels; superscripts r, s select ψ or d . Reversal transposes spin indices, exchanges types and endpoints, reverses momentum, and supplies a Grassmann minus sign. For lower indices the canonical discontinuity is $C^> - C^< = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}$ at equal times, in type-block notation. Conjugation provides an additional check: $C^{--,>}(k; 1, 2) = [C^{++,>}(k; 2, 1)]^\dagger$. No line includes a prefactor of i or a power of a .

2.3 Vertices and explicit index contraction

Write $J_+ = E$, $J_- = -E$. The two interaction insertions at a vertex on SK branch a are

$$-\frac{ia g_t a(\tau)}{2} \delta h(J_r)^{\alpha\beta} \zeta_{r\alpha} \zeta_{r\beta}, \quad r = +, -. \quad (18)$$

In explicit raised-index notation $J_+^{\alpha\beta} = -\epsilon^{\alpha\beta}$ and $J_-^{\dot{\alpha}\dot{\beta}} = +\epsilon^{\dot{\alpha}\dot{\beta}}$. Their numerical matrices are thus E and $-E$, respectively. Differentiating with respect to the two identical slots cancels $2!$ against $1/2$, giving the amputated rule $-ia g_t a(\tau) J_r$. Alternatively retain $1/2$ and sum all Wick slot assignments. These alternatives must not be combined. An external Higgs line supplies Eq. (7); integrate each vertex over $(-\infty, 0)$ with the BD prescription, and the independent loop momentum over $d^3 q / (2\pi)^3$.

For a cycle $i \rightarrow j \rightarrow k \rightarrow l \rightarrow i$ and fixed types, the explicit spin contraction is

$$\begin{aligned} & (J_{r_i})^{\alpha_i \beta_i} C_{ij, \beta_i \alpha_j}^{r_i r_j} (J_{r_j})^{\alpha_j \beta_j} C_{jk, \beta_j \alpha_k}^{r_j r_k} (J_{r_k})^{\alpha_k \beta_k} C_{kl, \beta_k \alpha_l}^{r_k r_l} (J_{r_l})^{\alpha_l \beta_l} C_{li, \beta_l \alpha_i}^{r_l r_i} \\ &= \text{tr}_2 [J_{r_i} C_{ij}^{r_i r_j} J_{r_j} C_{jk}^{r_j r_k} J_{r_k} C_{kl}^{r_k r_l} J_{r_l} C_{li}^{r_l r_i}]. \end{aligned} \quad (19)$$

This formula includes normal and anomalous lines with their correct index tensors, including words such as $++++$ and $+ - + -$. It is a sum of 2×2 traces, not an assignment of a conserved Dirac arrow to a Majorana line.

3 Connected box and its multiplicity

3.1 SK expansion and Majorana Wick counting

At fourth order expand $\exp[-ig_t \sum_a a \int a(\tau) \delta h(\psi\psi + \text{h.c.})/2]$. The $4!$ attachments of the labelled external Higgs fields cancel $1/4!$. Retain the four bilinear $1/2$ factors for the following

count. Of the $7!! = 105$ pairings of eight fermion slots, precisely 48 are connected without self-contractions. They form three undirected cycles, with 2^4 choices of which slot at each vertex connects to its next neighbor. Swapping these slots changes the Wick sign and the antisymmetric J sign together, so all 16 terms for a given cycle are equal. The remaining closed-cycle Wick permutation gives a minus sign. Thus $2^4/2^4 = 1$ for each *undirected* Majorana cycle, or $1/2$ for each of the six written orientations. This count is made before referring to a Dirac trace. A Pfaffian gives the same factor through $\log \text{Pf} = \frac{1}{2} \text{Tr} \log$.

Define the type-summed cycle tensor

$$\mathcal{T}_{ijkl} = \sum_{r_i, r_j, r_k, r_l = \pm} \text{tr}_2 [J_{r_i} C_{ij}^{r_i r_j} J_{r_j} C_{jk}^{r_j r_k} J_{r_k} C_{kl}^{r_k r_l} J_{r_l} C_{li}^{r_l r_i}]. \quad (20)$$

Take all external momenta incoming and choose

$$p_{ij} = q + k_i, \quad p_{jk} = q + k_i + k_j, \quad p_{kl} = q - k_l, \quad p_{li} = q, \quad (21)$$

where this equation is vectorial and each C_{uv} carries $(p_{uv}; \tau_u, \tau_v; a_u, a_v)$. For one Majorana copy and one color,

$$\begin{aligned} \mathcal{G}^{(1)} &= -\frac{g_t^4}{2} \sum_{(1,j,k,l)} \sum_{a_1, \dots, a_4 = \pm} \left(\prod_{v=1}^4 a_v \right) \int_{-\infty}^0 \prod_{v=1}^4 [d\tau_v a(\tau_v) G_{a_v}(k_v; \tau_v)] \int \frac{d^3 q}{(2\pi)^3} \mathcal{T}_{1jkl}, \\ \mathcal{G}^{\text{top}} &= 2N_c \mathcal{G}^{(1)}. \end{aligned} \quad (22)$$

The first sum is over permutations of $(j, k, l) = (2, 3, 4)$; the six cycles are 1234, 1243, 1324, 1342, 1423 and 1432. The loop sign and $(-i)^4 = 1$ are displayed separately from the SK weights. Equation (22) is the exact box expression. Appendix A proves its equality to the Dirac expression. Explicit eight-slot Wick enumeration in the code verifies that equality for all SK choices, rather than fixing the multiplicity by matching.

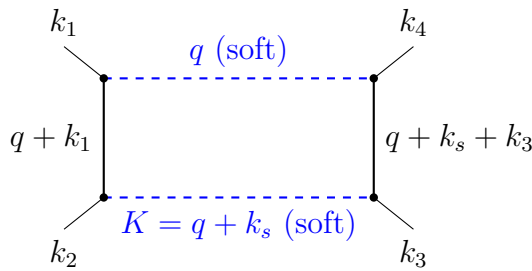


Figure 1: Representative 1234 cycle. All momentum labels are vectors. Dashed lines select the double-soft region; solid internal lines retain complete hard time dependence. Labels refer to the ordered contractions of Eq. (21), not to Majorana charge arrows.

3.2 Soft region and two-component mass projections

The double-soft factorization [3] selects four s -channel cycles as $\mathbf{k}_s = \mathbf{k}_1 + \mathbf{k}_2 \rightarrow 0$: 1234, 1243 near $q \sim k_s$, and 1342, 1432 near $q + k_1 \sim k_s$. The other two cycles have no double-soft s cut

for generic hard momenta. In 1234, the soft lines are q and $K = q + k_s$, while the right hard momentum is exactly $q - k_4 = q + k_s + k_3$. Dropping k_s from that line spoils the first nonzero coefficient. Overlapping collapsed channels are not assumed.

For reducing the type sum it is economical to form linear combinations of the 2×2 contractions. Define

$$N^{11} = C^{+-}, \quad N^{12} = C^{++}E, \quad N^{21} = -EC^{--}, \quad N^{22} = -EC^{-+}E, \quad b = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}. \quad (23)$$

These N^{rs} have one raised conjugate endpoint. Define the two-sector projections and their vertex weights by

$$R^{rs} = \sum_{u,v=1}^2 b_{ur} N^{uv} b_{vs}, \quad M^{rs} = \eta_r R^{rs}, \quad \eta_1 = 1, \quad \eta_2 = -1. \quad (24)$$

Here $r, s = 1, 2$ are projection labels, not the Majorana-copy label I . Every entry remains a 2×2 matrix. In these variables

$$\mathcal{T}_{ijkl} = \sum_{r_i, r_j, r_k, r_l=1}^2 \text{tr}_2 [M_{ij}^{r_i r_j} M_{jk}^{r_j r_k} M_{kl}^{r_k r_l} M_{li}^{r_l r_i}]. \quad (25)$$

The radial dictionary needed below follows by substituting the Whittaker modes, or equivalently Eq. (14):

	R^{11}	R^{12}	R^{21}	R^{22}
$>$	$F_1 F_2^* \mathbf{1}_2$	$F_1 G_2^* \sigma_k$	$G_1 F_2^* \sigma_k$	$G_1 G_2^* \mathbf{1}_2$
$<$	$-G_1^* G_2 \mathbf{1}_2$	$G_1^* F_2 \sigma_k$	$F_1^* G_2 \sigma_k$	$-F_1^* F_2 \mathbf{1}_2$

We abbreviate R^{rs} as R_{rs} when its other two indices explicitly carry SK labels; $\sigma_k = \boldsymbol{\sigma} \cdot \hat{k}$. These projections permit reuse of scalar integrals without making four-component modes the main calculation.

4 Collapsed expansion and cancellation

4.1 Nonanalytic soft branches

For $x = -k\tau \ll 1$, introduce

$$A = -\frac{i2^{-1/2-i\mu}}{\sqrt{\pi}} \Gamma(\tfrac{1}{2} - i\mu), \quad c = -A^2 = \frac{2^{-1-2i\mu}}{\pi} \Gamma(\tfrac{1}{2} - i\mu)^2, \quad |c| = \frac{1}{2 \cosh \pi\mu}. \quad (26)$$

The radial expansions are

$$\begin{aligned} F(x) &= e^{\pi\mu/2} A x^{i\mu} - \frac{i e^{-\pi\mu/2} A^*}{1 - 2i\mu} x^{1-i\mu} + \dots, \\ G(x) &= -e^{-\pi\mu/2} A^* x^{-i\mu} + \frac{i e^{\pi\mu/2} A}{1 + 2i\mu} x^{1+i\mu} + \dots. \end{aligned} \quad (27)$$

The leading nonanalytic projections are $R_{\text{NL}}^{12} = c(k^2 \tau \tau')^{i\mu} \sigma_k$ and $R_{\text{NL}}^{21} = c^*(k^2 \tau \tau')^{-i\mu} \sigma_k$; the zeroth-order diagonal terms depend only on τ/τ' and are analytic in $\mathbf{k}$. The nonanalytic pieces, including

descendants, agree in $>$ and the *signed* $<$ contraction. This follows from the two Bessel branches of Eq. (13), or from the reversal identity in Eq. (16); keeping the same momentum on a reversed Majorana line would destroy this identity. They therefore have no SK labels. Analytic soft terms do not generate the double-soft nonanalytic coefficient; hard-region contributions supply contact terms and their matching, excluded here.

The entire $+i\mu$ soft branch can be written in 2×2 factors:

$$\begin{aligned} M_{\text{NL},+}^{rs}(k; \tau, \tau') &= c(k^2 \tau \tau')^{i\mu} L_r(k, \tau) \sigma_k U_s(k, \tau'), \\ L_1 &= U_2 = f(x) \mathbf{1}_2, \quad L_2 = U_1 = g(x) \sigma_k, \\ f(x) &= {}_0F_1\left(\frac{1}{2} + i\mu; -x^2/4\right), \quad g(x) = -\frac{ix}{1 + 2i\mu} {}_0F_1\left(\frac{3}{2} + i\mu; -x^2/4\right). \end{aligned} \quad (28)$$

Thus $L_2 = i\tau \boldsymbol{\sigma} \cdot \mathbf{k} / (1 + 2i\mu) + O(k^3 \tau^3)$, while $L_1 = \mathbf{1}_2 + O(k^2 \tau^2)$, and likewise for U . The conjugate branch is determined by $R_{\text{NL},-}(k; \tau, \tau') = [R_{\text{NL},+}(k; \tau', \tau)]^\dagger$ in sector-block notation, before the η_r vertex weights are applied. This exact ${}_0F_1$ form also checks that no descendant of the required order has been omitted.

4.2 Bose exchange and the first surviving hard blocks

We now extract the nonanalytic signal from the four s -cut cycles. All formulas from this point include both Majorana copies: the factor $2N_c$ cancels the $1/2$ assigned to each written cycle in Eq. (22). For $k_s \ll k_L, k_R$, use symmetric hard momenta:

$$\mathbf{k}_1 = \mathbf{k}_L + \frac{1}{2} \mathbf{k}_s, \quad \mathbf{k}_2 = -\mathbf{k}_L + \frac{1}{2} \mathbf{k}_s, \quad (29)$$

$$\mathbf{k}_3 = \mathbf{k}_R - \frac{1}{2} \mathbf{k}_s, \quad \mathbf{k}_4 = -\mathbf{k}_R - \frac{1}{2} \mathbf{k}_s, \quad (30)$$

$$\boldsymbol{\ell} = \mathbf{q} + \frac{1}{2} \mathbf{k}_s, \quad \mathbf{K} = \mathbf{q} + \mathbf{k}_s, \quad \mathbf{n}_L = \hat{\mathbf{k}}_L, \quad \mathbf{n}_R = \hat{\mathbf{k}}_R, \quad \mathbf{n}_s = \hat{\mathbf{k}}_s. \quad (31)$$

Write $k_L = |\mathbf{k}_L|$, $k_R = |\mathbf{k}_R|$; at the accuracy of the final answer $k_L = k_{12}/2$, $k_R = k_{34}/2$, where $k_{ij} = k_i + k_j$ is a sum of magnitudes.

For completeness the four routed cycles, cyclically started at a left hard line, are

cycle	left hard	soft K	right hard	soft q
1234	12 : $k_L + \ell$	23	34 : $k_R + \ell$	41
1243	12 : $k_L + \ell$	24	43 : $-k_R + \ell$	31
1342	21 : $-k_L + \ell$	13	34 : $k_R + \ell$	42
1432	21 : $-k_L + \ell$	14	43 : $-k_R + \ell$	32

All momenta in this table are vectors. The last two rows follow from shifting the original routing variable to $q_{\text{old}} = K - k_1$. It follows directly that their sum is the product of the two independent hard-end ordering sums, after relabelling each pair's dummy times and SK labels. No equality of unequal external magnitudes is assumed here.

Each hard line is contracted with the endpoint factors of its two soft neighbors. For the representative cycle these are the 2×2 matrices

$$V_L = \sum_{r,s=1}^2 U_r(q, \tau_1) M_{12}^{rs}(\mathbf{k}_L + \boldsymbol{\ell}; \tau_1, \tau_2) L_s(K, \tau_2), \quad (32)$$

$$V_R = \sum_{r,s=1}^2 U_r(K, \tau_3) M_{34}^{rs}(\mathbf{k}_R + \boldsymbol{\ell}; \tau_3, \tau_4) L_s(q, \tau_4). \quad (33)$$

The type and projection sums in the box therefore reduce to

$$c^2(qK)^{2i\mu} \prod_v (-\tau_v)^{i\mu} \text{tr}_2[V_L \sigma_K V_R \sigma_q]. \quad (34)$$

The remaining spin trace reduces to scalar products through

$$\text{tr}[(\boldsymbol{\sigma} \cdot \mathbf{A}) \sigma_K (\boldsymbol{\sigma} \cdot \mathbf{B}) \sigma_q] = 2[(\mathbf{A} \cdot \hat{K})(\mathbf{B} \cdot \hat{q}) + (\mathbf{A} \cdot \hat{q})(\mathbf{B} \cdot \hat{K}) - (\mathbf{A} \cdot \mathbf{B})(\hat{K} \cdot \hat{q})]. \quad (35)$$

At zeroth order only $U_2 = L_1 = \mathbf{1}_2$ survives. These projections select $V_L^{(0)} = -R_{21}(k_L) \propto \boldsymbol{\sigma} \cdot \mathbf{n}_L$, and similarly $V_R^{(0)} \propto \boldsymbol{\sigma} \cdot \mathbf{n}_R$.

The four cut cycles are generated from 1234 by the independent exchanges $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$, giving 1234, 1342, 1243, 1432 after cyclic relabelling. The left exchange sends $\mathbf{n}_L \rightarrow -\mathbf{n}_L$ at fixed $\mathbf{k}_s$; the right sends $\mathbf{n}_R \rightarrow -\mathbf{n}_R$. At this order $k_1 = k_2 = k_L$ and $k_3 = k_4 = k_R$. After relabelling the dummy vertex times and SK indices, the scalar time kernels are the same for both orderings, whereas each corresponding Pauli matrix changes sign. Thus the zeroth-order four-cycle spin factor is proportional to

$$\sum_{\varepsilon_L, \varepsilon_R = \pm 1} \varepsilon_L \varepsilon_R \text{tr}_2[(\boldsymbol{\sigma} \cdot \mathbf{n}_L) \sigma_K (\boldsymbol{\sigma} \cdot \mathbf{n}_R) \sigma_q] = 0.$$

This is a cancellation between the external-leg orderings, not a vanishing trace for an individual box. It holds after relabelling the hard time integrals but before the soft momentum integral is evaluated. Individual projected hard SK sums are nonzero; this is distinct from the late-time SK cancellation discussed in Eq. (66). Thus

$$\boxed{\mathcal{G}_{s,\text{clock}}^{\text{box}} \Big|_{k_s^{3 \pm 4i\mu}} = 0.} \quad (36)$$

Keeping a first-order correction at only one hard end still leaves a vanishing zeroth-order sum at the other. The first surviving clock term therefore needs one soft-momentum correction at *each* end: both the endpoint corrections in Eq. (28) and the momentum dependence of the hard propagators must be retained. Their product supplies two extra powers of k_s , changing the clock scaling to $k_s^{5 \pm 4i\mu}$. The omitted second-order endpoint terms multiply a vanishing zeroth-order hard block at the opposite end and cannot change this leading coefficient. The next subsection evaluates the two first-order hard blocks explicitly. This cancellation concerns the equal soft branches; the opposite-branch, nonoscillatory contribution is treated separately in Sec. 6.2.

4.3 Left and right hard subgraphs

We now calculate the first order of each Bose-symmetrized hard block. Write $x = -k_L \tau_1$, $y = -k_L \tau_2$ and $r = p/k_L$, where $p = |\mathbf{p}|$ denotes the hard-line momentum magnitude in this section. Also write $\sigma_p = \boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$. We need only two SK-summed scalar seeds, corresponding to the 21 and 11 blocks of R :

$$\mathcal{H}_{mn}(\alpha, \beta; r) = \sum_{a,b=\pm} ab \int_0^\infty dx dy x^{\alpha-1} y^{\beta-1} (1 + iax)(1 + iby) e^{-iax - iby} [R_{mn}(r; x, y)]_{ab}^{\text{radial}},$$

$$mn = 21, 11. \quad (37)$$

“Radial” means the coefficient of σ_p in R_{21} or of $\mathbf{1}_2$ in R_{11} ; their mode arguments are rx, ry . Hence these are scalar integrals, with the spin structure already removed. Their finite evaluation is given in Sec. 5.1.

The useful exact reversal identities are

$$R_{21;ab}(\tau_1, \tau_2) = R_{21;ba}(\tau_2, \tau_1), \quad R_{11;ab}(\tau_1, \tau_2) = -R_{22;ba}(\tau_2, \tau_1). \quad (38)$$

For one left ordering, expanding the endpoint projectors gives

$$V_L = -R_{21} + \frac{i\tau_1}{1+2i\mu} \boldsymbol{\sigma} \cdot \mathbf{q} R_{11} - \frac{i\tau_2}{1+2i\mu} R_{22} \boldsymbol{\sigma} \cdot \mathbf{K} + O(q^2, k_s^2). \quad (39)$$

All entries here are unapproximated hard SK contractions before expansion of their momentum $\mathbf{p} = \mathbf{k}_L + \boldsymbol{\ell}$. The other ordering has $\mathbf{p} = -\mathbf{k}_L + \boldsymbol{\ell}$. The derivative of the hard line and the descendants of both soft endpoints must both be kept. Eq. (38) shows that the first-order difference between the two external magnitudes gives zero in the sum: the integrated R_{21} block is symmetric in those energies. Also $\int \tau_2 R_{22} = -\int \tau_1 R_{11}$ at equal external magnitudes.

The intermediate momentum expansion makes the two terms in Eq. (46) explicit. Write $h(r) = \mathcal{H}_{21}(i\mu, i\mu; r)$ and $L = \mathbf{n}_L \cdot \boldsymbol{\ell}$. For the ordering $\varepsilon = \pm 1$,

$$|\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}| = k_L + \varepsilon L + O(\ell^2/k_L), \quad (40)$$

$$\boldsymbol{\sigma} \cdot \widehat{\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}} = \varepsilon \boldsymbol{\sigma} \cdot \mathbf{n}_L + \frac{\boldsymbol{\sigma} \cdot (\boldsymbol{\ell} - \mathbf{n}_L L)}{k_L} + O(\ell^2/k_L^2), \quad (41)$$

$$h\left(\frac{|\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}|}{k_L}\right) = h(1) + \frac{\varepsilon L}{k_L} h'(1) + O(\ell^2/k_L^2). \quad (42)$$

The $-R_{21}$ terms of the two orderings therefore add to

$$-\frac{2}{k_L} \boldsymbol{\sigma} \cdot [h(1)\boldsymbol{\ell} + (h'(1) - h(1))\mathbf{n}_L L] + O(\ell^2/k_L^2). \quad (43)$$

The R_{11} and R_{22} descendants in Eq. (39) contribute $-4i\mathcal{H}_{11}(1+i\mu, i\mu; 1)\boldsymbol{\sigma} \cdot \boldsymbol{\ell}/[k_L(1+2i\mu)]$. The common factor $k_L^{-2i\mu}$ comes from $(-\tau_1)^{i\mu}(-\tau_2)^{i\mu}$ after $x = -k_L\tau_1$, $y = -k_L\tau_2$. Finally, the external magnitudes are $|\mathbf{k}_L \pm \mathbf{k}_s/2| = k_L \pm \mathbf{n}_L \cdot \mathbf{k}_s/2 + O(k_s^2/k_L)$. Their first-order shifts cancel under the two orderings by the time-reversal identity in Eq. (38); equal external energies are imposed only after this sum.

Strip $H^2/(4k_1^3 k_2^3)$ from the left two external lines and vertices. Its integrated projected matrix, summed over its two Higgs orderings, is

$$\mathcal{V}_L = -2k_L^{-1-2i\mu} \boldsymbol{\sigma} \cdot [\mathcal{P}_1(\mu)\boldsymbol{\ell} + \mathcal{P}_2(\mu)\mathbf{n}_L(\mathbf{n}_L \cdot \boldsymbol{\ell})], \quad (44)$$

$$\mathcal{P}_1(\mu) = \mathcal{H}_{21}(i\mu, i\mu; 1) + \frac{2i}{1+2i\mu} \mathcal{H}_{11}(1+i\mu, i\mu; 1), \quad (45)$$

$$\mathcal{P}_2(\mu) = (\partial_r - 1)\mathcal{H}_{21}(i\mu, i\mu; r)|_{r=1}. \quad (46)$$

The two coefficients $\mathcal{P}_1, \mathcal{P}_2$ multiply the independent spin structures $\boldsymbol{\sigma} \cdot \boldsymbol{\ell}$ and $(\boldsymbol{\sigma} \cdot \mathbf{n}_L)(\mathbf{n}_L \cdot \boldsymbol{\ell})$. The $\mathcal{H}_{11}$ term comes from the soft endpoint descendants; the r derivative comes from the hard

propagator's momentum dependence. The right result is the same with $(k_L, \mathbf{n}_L) \rightarrow (k_R, \mathbf{n}_R)$ and external prefactor $H^2/(4k_3^3 k_4^3)$. In particular the four-cycle sum is the product of these two already symmetrized blocks; no further factor of two or four is allowed. Formula (46) retains all terms at this order. The absence of the constant term in each block makes the signal a spatial P -wave fermion-pair effect. It adds two powers of k_s to the bubble scaling.

5 Scalar time integrals and soft convolution

5.1 Evaluation of the scalar hard seeds

Relation to the published Hankel scalar seed. The hard time integrals can also be reduced to the scalar seed of Ref. [2], provided its mass index is analytically continued. This gives a direct route to the literature's single- and double-folded formulae. Set

$$\nu_h = \mu - \frac{i}{2}, \quad \rho = i\nu_h = \frac{1}{2} + i\mu, \quad x = -k_L\tau_1, \quad y = -k_L\tau_2, \quad r = p/k_L. \quad (47)$$

Define a dimensionless scalar SK kernel $S_{ab}^{(\nu_h)}$ from the two Wightman functions

$$S^>(r; x, y) = \frac{\pi}{4} e^{-\pi\nu_h} (xy)^{3/2} H_\rho^{(1)}(rx) H_{-\rho}^{(2)}(ry), \quad (48)$$

$$S^<(r; x, y) = \frac{\pi}{4} e^{-\pi\nu_h} (xy)^{3/2} H_{-\rho}^{(2)}(rx) H_\rho^{(1)}(ry), \quad (49)$$

with the same SK theta-function ordering as Eq. (17). Using $H_{-\rho}^{(2)} = e^{-i\pi\rho} H_\rho^{(2)}$ and $N_F N_G = \pi/4$ for the mode prefactors in Eq. (13), one finds the *exact* radial identities, for every branch pair,

$$[R_{21}(r; x, y)]_{ab}^{\text{radial}} = \frac{r}{xy} S_{ab}^{(\nu_h)}(r; x, y), \quad [R_{11}(r; x, y)]_{ab}^{\text{radial}} = \frac{1}{r} \left(-i\partial_x + \frac{\mu}{x} \right) [R_{21}(r; x, y)]_{ab}^{\text{radial}}. \quad (50)$$

The $1/r$ in the second identity follows from differentiating a mode with argument rx . Differentiating the SK theta functions introduces no delta-function term: the two radial R_{21} Wightman functions agree at $x = y$. Although ν_h is complex, Eq. (49) is an analytic scalar kernel; no physical scalar with complex mass is being introduced.

To see precisely which published seeds are needed, let

$$J_{ab}^{p,q}(r) = \int_0^\infty dx dy x^p y^q e^{-iax - iby} S_{ab}^{(\nu_h)}(r; x, y). \quad (51)$$

In the units $H = 1$ of Ref. [2], its branch-resolved Hankel seed $\mathcal{J}_{ab}^{p,q}$ obeys the exact normalization map

$$J_{ab}^{p,q}(r) = -ab r^{-5-p-q} \mathcal{J}_{ab}^{p,q}(u, u), \quad u = \frac{2r}{1+r}. \quad (52)$$

The reference includes $-ab$ in the seed definition; neither that sign nor the power of r may be omitted. For $p = \alpha - 2$, $q = \beta - 2$, Eq. (37) becomes

$$\mathcal{H}_{21}(\alpha, \beta; r) = r \sum_{a,b=\pm} ab \left[J_{ab}^{p,q} + ia J_{ab}^{p+1,q} + ib J_{ab}^{p,q+1} - ab J_{ab}^{p+1,q+1} \right]. \quad (53)$$

The four terms correspond to the integer time-power shifts generated by $(1 + iax)(1 + iby)$. The SK indices stay explicit until their a, b weights are combined.

The remaining hard coefficient can also be expressed without a new time integral. Put $\mathcal{S}_{ab}^{p,q} = \mathcal{S}_{ab}^{p,q}(1, 1)$, $p_0 = -2 + i\mu + \delta$, and use one common endpoint regulator δ in all seed powers. Equation (53) gives

$$\mathcal{H}_{21}(i\mu + \delta, i\mu + \delta; 1) = - \sum_{a,b=\pm} \sum_{j,l=0}^1 (ia)^j (ib)^l \mathcal{S}_{ab}^{p_0+j, p_0+l}. \quad (54)$$

Integrating the second relation of Eq. (50) by parts yields

$$\begin{aligned} \mathcal{H}_{11}(1 + i\mu + \delta, i\mu + \delta; 1) &= i\delta \mathcal{H}_{21}(i\mu + \delta, i\mu + \delta; 1) \\ &\quad - i \sum_{a,b=\pm} \left[\mathcal{S}_{ab}^{i\mu+\delta, p_0} + ib \mathcal{S}_{ab}^{i\mu+\delta, p_0+1} \right]. \end{aligned} \quad (55)$$

The $i\delta \mathcal{H}_{21}$ term is retained at finite regulator; the complete $\mathcal{H}_{21}$ will be finite, so this term vanishes in the final limit. Individual SK terms do have regulator poles. The integration-by-parts boundary term at $x = 0$ cancels only in the complete SK sum, since for $x < y$ the two branches with the same b have the same Wightman function.

The hard-momentum derivative in $\mathcal{P}_2$ follows from the published scalar bootstrap equation without differentiating a time integral. At $u_1 = u_2 = 1$, the coefficient of its second derivative vanishes. Define

$$T_{ab}^{p,q} = \delta_{ab} e^{-ia\pi(p+q)/2} 2^{-5-p-q} \Gamma(5 + p + q), \quad (56)$$

$$\mathcal{D}_{ab}^{p,q} = \frac{1}{2} \sum_{w=p,q} \frac{\nu_h^2 + (w + 5/2)^2}{w + 3} \mathcal{S}_{ab}^{p,q} - \frac{1}{2} \left(\frac{1}{p+3} + \frac{1}{q+3} \right) T_{ab}^{p,q}. \quad (57)$$

Since $du/dr|_{r=1} = 1/2$, $\mathcal{D}_{ab}^{p,q}$ is precisely $d\mathcal{S}_{ab}^{p,q}(u(r), u(r))/dr|_{r=1}$. Combining this with the explicit r^{-5-p-q} in Eq. (52) gives

$$\mathcal{P}_2 = -\text{FP}_{\delta \rightarrow 0} \sum_{a,b=\pm} \sum_{j,l=0}^1 (ia)^j (ib)^l \left[\mathcal{D}_{ab}^{p_0+j, p_0+l} - (5 + 2p_0 + j + l) \mathcal{S}_{ab}^{p_0+j, p_0+l} \right]. \quad (58)$$

Equations (54)–(58), with $\mathcal{P}_1$ from Eq. (46), complete both hard subgraphs in terms of the scalar seed values of Ref. [2]; the right subgraph uses the same two functions.

Regularization and a finite evaluation. There is an essential normalization in Eq. (92) of Ref. [2]: its hypergeometric function is regularized,

$${}_3\tilde{F}_2(a, b, c; d, e; 1) = \frac{{}_3F_2(a, b, c; d, e; 1)}{\Gamma(d)\Gamma(e)}. \quad (59)$$

A historical implementation omitted these two Gamma factors; the workspace version preceding this rewrite had already corrected them. This is not a new correction from the Majorana change of variables. A real-index conjugation check alone cannot detect this error. As an independent normalization check, the conformal scalar $\nu_h = -i/2$ has $S^>(1; x, y) = xy e^{i(x-y)}/2$. Direct elementary time integration gives $\mathcal{S}_{++}^{0,0} = \mathcal{S}_{--}^{0,0} = -3/16$ and $\mathcal{S}_{+-}^{0,0} = \mathcal{S}_{-+}^{0,0} = 1/32$; the corrected implementation reproduces these values and their unequal-power generalization.

The scalar kernel is invariant under $\nu_h \rightarrow -\nu_h$, by the Hankel order-reflection identities. In each equal-SK seed choose the sign of ν_h so that $aiv_h = 1/2 + i\mu$. Its ordinary ${}_3F_2(1)$ then has convergence balance $1 + i\mu$, whose real part is positive. This is an exact change of representation and needs no additional mass-index regulator.

For this subsection only set $z = i\mu$, and define the single symmetric family

$$\mathcal{Q}_{jl} = \frac{2^{-1-2z-j-l}\Gamma(1+2z+j+l)}{(1+2z+j)(1+2z+l)} {}_3F_2\left(\begin{matrix} 1+2z+j+l, 1+z, 1 \\ 2+2z+j, 2+2z+l \end{matrix}; 1\right). \quad (60)$$

Only $\mathcal{Q}_{00}, \mathcal{Q}_{01}, \mathcal{Q}_{11}, \mathcal{Q}_{02}, \mathcal{Q}_{12}$ are needed. Also abbreviate the finite endpoint contribution by

$$\mathcal{E} = \pi^2 2^{-1-2z} \frac{\Gamma(1+2z)^2}{\Gamma(1+z)^2}. \quad (61)$$

The fully summed hard seeds and momentum derivative are

$$\begin{aligned} \mathcal{H}_{21}(z, z; 1) &= 2 \cosh(\pi\mu)(\mathcal{Q}_{00} + 2\mathcal{Q}_{01} + \mathcal{Q}_{11}) - \mathcal{E}, \\ \mathcal{H}_{11}(1+z, z; 1) &= -2i \cosh(\pi\mu)(\mathcal{Q}_{02} + \mathcal{Q}_{12}), \\ \mathcal{P}_2 &= 2 \cosh(\pi\mu) \sum_{j,l=0}^1 \left[\frac{j(1+2z+j)}{2(1+z+j)} + \frac{l(1+2z+l)}{2(1+z+l)} - (1+2z+j+l) \right] \mathcal{Q}_{jl} \\ &\quad + (1+2z)\mathcal{E} + 2^{-1-2z} \cosh(\pi\mu)\Gamma(1+2z) \frac{4+5z+2z^2}{1+z}. \end{aligned} \quad (62)$$

Together with $\mathcal{P}_1$ in Eq. (46), these are finite, convergent special-function expressions for the complete hard coefficients. No raw mode-power-series truncation or numerical regulator extrapolation is required for the oscillatory signal.

Here is the endpoint cancellation leading to these expressions. For $p_j = -2 + z + \delta + j$, write the Gamma product in the opposite-SK seed as

$$C_{jl}(\delta) = \frac{\Gamma(\delta+j)\Gamma(1+2z+\delta+j)\Gamma(\delta+l)\Gamma(1+2z+\delta+l)}{\Gamma(1+z+\delta+j)\Gamma(1+z+\delta+l)}. \quad (64)$$

After the external-mode factors $(ia)^j(ib)^l$ and the SK branches are summed, its contribution to $\mathcal{H}_{21}$ is $2 \cdot 2^{-1-2z-2\delta-j-l} [\cos(\pi\delta) - 1] C_{jl}(\delta)$. Only $j = l = 0$ has a double pole; therefore

$$\lim_{\delta \rightarrow 0} 2[\cos(\pi\delta) - 1] C_{00}(\delta) = -\pi^2 \frac{\Gamma(1+2z)^2}{\Gamma(1+z)^2}. \quad (65)$$

This is precisely $-\mathcal{E}$ after restoring its power of two. Discarding the poles branch by branch would lose this finite term. The remaining regularized-hypergeometric pieces give $2 \cosh(\pi\mu) \mathcal{Q}_{jl}$. The integration-by-parts insertion in Eq. (55) replaces $j = 0, 1$ by $j = 2$ and contributes $-i$ times the same pieces; it has no double-pole remainder. Finally, substituting p_j in Eq. (57) gives the bracket in Eq. (63). The last term in that equation is the summed bootstrap source, and $(1+2z)\mathcal{E}$ is the derivative's endpoint remainder. These steps preserve all factors of r .

The late-time cancellation can also be seen before integration. With $E_a(x) = (1 + iax)e^{-iax}$, the SK sum for $x < y$ is

$$\sum_{a,b} ab E_a(x) E_b(y) R_{ab}(x, y) = [E_+(x) - E_-(x)][E_+(y) R^>(x, y) - E_-(y) R^<(x, y)]. \quad (66)$$

Since $E_+(x) - E_-(x) = -2ix^3/3 + O(x^5)$, the common late-time endpoint is integrable in the hard blocks used here. The early-time BD prescription is still necessary; this identity does not justify integrating undamped oscillations numerically on a truncated interval.

For reference, direct evaluation of Eqs. (62) and (63) gives

μ	$\mathcal{P}_1$	$\mathcal{P}_2$
0.20	$-1.030411081 - 1.109020094i$	$3.163771501 + 0.704629904i$
0.73	$-1.598130708 - 2.346219694i$	$1.923885009 + 1.958596831i$
1.00	$-1.325125325 - 2.665004109i$	$1.137292037 + 2.131408832i$
2.00	$1.901463897 - 2.535666708i$	$-1.119241948 + 1.383844867i$
5.00	$3.921820606 + 1.458261456i$	$-2.047325876 - 0.690483918i$

The corrected branch-resolved calculation converges to these values with a common endpoint regulator. The two representations are checked in `checks/weyl_seed_checks.py`; the physical coefficients are implemented in `checks/weyl_scalar_seed.py`.

5.2 Soft bubble integral

The spin trace of Eq. (35) leaves a polynomial numerator divided by qK . To display it, write the two vectors appearing in the hard blocks as

$$\mathbf{v}_L = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_L (\mathbf{n}_L \cdot \boldsymbol{\ell}), \quad \mathbf{v}_R = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_R (\mathbf{n}_R \cdot \boldsymbol{\ell}). \quad (67)$$

Since $\mathbf{K} = \boldsymbol{\ell} + \mathbf{k}_s/2$ and $\mathbf{q} = \boldsymbol{\ell} - \mathbf{k}_s/2$, the scalar loop integral after the spin contraction is

$$2 \int \frac{d^3 q}{(2\pi)^3} (qK)^{2i\mu-1} \left[2(\mathbf{v}_L \cdot \boldsymbol{\ell})(\mathbf{v}_R \cdot \boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L \cdot \mathbf{k}_s)(\mathbf{v}_R \cdot \mathbf{k}_s) - (\mathbf{v}_L \cdot \mathbf{v}_R)(\ell^2 - k_s^2/4) \right]. \quad (68)$$

Its mass dimension is $5 + 4i\mu$. We now evaluate it by scalar convolutions, following the same procedure as for the vector box. The only required scalar formula is

$$F_0(A, B) = \frac{\Gamma(A + B - \frac{3}{2})\Gamma(\frac{3}{2} - A)\Gamma(\frac{3}{2} - B)}{\Gamma(A)\Gamma(B)\Gamma(3 - A - B)}, \quad (69)$$

$$\int \frac{d^3 q}{(2\pi)^3} (q^2)^{-A} (K^2)^{-B} = \frac{k_s^{3-2A-2B}}{(4\pi)^{3/2}} F_0(A, B). \quad (70)$$

For the scalar numerator this identity follows by Schwinger parameters and a Gaussian momentum integral in the convergence domain $\Re A, \Re B < 3/2$, $\Re(A + B) > 3/2$. The shifted polynomial moments are defined by the same meromorphic continuation, with common regulator parameters before recombination. The physical soft-region integrals have UV power divergences when extended beyond $q \sim k_s$; this continuation selects their nonanalytic coefficient. UV-dependent polynomial contact terms belong to matching with the hard region and are excluded. No physical UV cutoff is sent to infinity term by term.

For the reduction only, measure q, K, ℓ in units of k_s , put $u = q^2$, $v = K^2$, and set $\nu = 1/2 - i\mu$. Then

$$\ell^2 = (u + v)/2 - 1/4, \quad \ell_{\parallel} \equiv \boldsymbol{\ell} \cdot \mathbf{n}_s = (v - u)/2, \quad \ell_{\perp}^2 = \ell^2 - \ell_{\parallel}^2. \quad (71)$$

Azimuthal averaging replaces two transverse components by $\ell_\perp^2 \delta_{ij}^\perp/2$ and four by $\ell_\perp^4 (\delta_{ij}^\perp \delta_{kl}^\perp + \delta_{ik}^\perp \delta_{jl}^\perp + \delta_{il}^\perp \delta_{jk}^\perp)/8$, where $\delta_{ij}^\perp = \delta_{ij} - n_{si} n_{sj}$. The numerator in Eq. (68) consequently becomes a polynomial in u, v and the three external angles. Each monomial is integrated by

$$u^m v^n \longmapsto \frac{F_0(\nu - m, \nu - n)}{(4\pi)^{3/2}}, \quad (72)$$

$$\frac{F_0(\nu - m, \nu - n)}{F_0(\nu, \nu)} = \frac{(3/2 - \nu)_m (3/2 - \nu)_n (1 - \nu)_m (1 - \nu)_n}{(5/2 - 2\nu)_{m+n} (3 - 2\nu)_{m+n}}. \quad (73)$$

Only nonnegative integers m, n with $m + n \leq 4$ occur. Thus the bubble is a finite Gamma sum, and all its terms reduce to the same $F_0(\nu, \nu)$ times rational functions. For example the coefficient of $\mathcal{P}_1^2$ in the square bracket of Eq. (68) is $\ell^4 + \ell^2/4 - \ell_\parallel^2/2$, whose integral is

$$\frac{F_0(\nu, \nu)}{(4\pi)^{3/2}} \frac{3(\nu - 1)(2\nu - 3)}{8(\nu - 2)(4\nu - 7)(4\nu - 5)}. \quad (74)$$

The $\mathcal{P}_1 \mathcal{P}_2$ and $\mathcal{P}_2^2$ terms are evaluated by the same two steps. Their fully reduced coefficients are displayed directly in Eq. (82), without introducing separate symbols for each tensor moment. The azimuthal average introduces only the five angle monomials $1, c_1^2 + c_3^2, c_1^2 c_3^2, \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3$, and $\sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3)$, where $c_i = \cos \theta_i$. Applying Eq. (73) to their coefficients reproduces Eq. (82).

6 Nonlocal signal

6.1 Oscillatory signal and angular coefficients

Choose the angle convention

$$\mathbf{n}_L = (\sin \theta_1, 0, \cos \theta_1), \quad \mathbf{n}_R = (\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3), \quad \mathbf{n}_s = (0, 0, 1). \quad (75)$$

The leading nonzero oscillatory term, summed over the four s -cut boxes, is

$$\begin{aligned} \langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle' |_{s, \text{clock}} = & - \frac{N_c g_t^4 H^4 k_s^5}{16 k_1^3 k_2^3 k_3^3 k_4^3 k_L k_R} \\ & \times \left[\left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2i\mu} \mathcal{C}(\mu; \theta_1, \theta_3, \phi_3) + \text{c.c.} \right]. \end{aligned} \quad (76)$$

The coefficient is given directly in the angular basis of the vector-box note:

$$\begin{aligned} \mathcal{C} = & A_1 + A_2(\cos^2 \theta_1 + \cos^2 \theta_3) + A_3 \cos^2 \theta_1 \cos^2 \theta_3 \\ & + B_1 \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3 + D_1 \sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3). \end{aligned} \quad (77)$$

All five coefficients share one soft-bubble normalization, including the two soft-mode coefficients c^2 , the factor 4 from the two Bose-summed hard blocks, the factor 2 from the Pauli trace, and the phase from the chosen momentum ratio:

$$\mathcal{B}(\mu) = \frac{2^{1+4i\mu} \Gamma(\frac{1}{2} - i\mu)^4 F_0(\nu, \nu)}{32\pi^2 (4\pi)^{3/2} (\nu - 3)(\nu - 2)(4\nu - 7)(4\nu - 5)}, \quad \nu = \frac{1}{2} - i\mu. \quad (78)$$

In particular, the natural hard scaling is $(k_s^2/(k_L k_R))^{2i\mu}$. Since $4k_{12}k_{34} = 16k_L k_R$ at this order, using the vector note's ratio transfers $16^{2i\mu}$ into the coefficient. Thus $8c^2 16^{2i\mu} = 2^{1+4i\mu} \Gamma(1/2 - i\mu)^4/\pi^2$, which fixes both the phase and the overall factor in Eq. (78). Combining the bubble with the two hard coefficients of Eq. (46) gives the final mass dependence:

$$A_1 = \mathcal{B}(\nu - 1)(2\nu - 3) \left[4(\nu - 3)(3\mathcal{P}_1^2 + 2\mathcal{P}_1\mathcal{P}_2) + (2\nu - 5)\mathcal{P}_2^2 \right], \quad (79)$$

$$A_2 = -\mathcal{B}\mathcal{P}_2^2(\nu - 1)(2\nu - 3)^2, \quad A_3 = -3A_2, \quad (80)$$

$$B_1 = -2\mathcal{B}\mathcal{P}_2^2(\nu - 1)(\nu - 2)(2\nu - 3), \quad (81)$$

$$D_1 = \mathcal{B}\mathcal{P}_2^2(\nu - 2)(2\nu - 5)(2\nu - 3). \quad (82)$$

Thus the answer requires only two hard functions $\mathcal{P}_1, \mathcal{P}_2$ and one explicit bubble factor $\mathcal{B}$. The coefficients are complex and include the clock phase. The full answer is even under either Higgs-pair exchange and under parity. Its engineering dimension is -5 , as required for the primed four-point function. The powers of H follow from four rescaled Weyl Yukawa vertices and four external Higgs contractions; no extra factor of H or a second color/orientation multiplicity is implicit. For fixed $\mu > 0$ these are the leading double-soft nonanalytic terms; higher soft powers and analytic contact terms are not displayed. With the centered hard momenta the next clock terms begin at $k_s^{7\pm 4i\mu}$, and the next mixed terms at k_s^5 . Thus the generic relative error within each branch is $O(k_s^2/k_L^2, k_s^2/k_R^2, k_s^2/(k_L k_R))$; it is not a relative-error bound at a zero of the displayed angular coefficient.

6.2 Nonoscillatory contribution

Taking opposite soft branches gives a nonoscillatory k_s^3 term. The hard blocks are R_{11} and $-R_{22}$; Eq. (38) makes their integrated values equal to the real seed $\mathcal{H}_{11}(-i\mu, +i\mu; 1)$. The leading soft matrices are now opposite triangular blocks, so they project the diagonal hard blocks. Each hard end therefore has a Bose-even zeroth-order term. There are two soft assignments and two Higgs orderings at each end, giving $2 \times 2 \times 2$ contributions; the spin trace adds $2\hat{q} \cdot \hat{K}$. The soft coefficients give $|c|^2 = 1/(4 \cosh^2 \pi\mu)$. These factors fix the normalization of the mixed term independently of the oscillatory branch. The scalar bubble in this case is

$$\int \frac{d^3 q}{(2\pi)^3} (q^2)^{-i\mu} (K^2)^{i\mu} \hat{q} \cdot \hat{K} = \frac{k_s^3}{(4\pi)^{3/2}} \frac{2\sqrt{\pi}}{3} \mu(1 + \mu^2) \coth(\pi\mu). \quad (83)$$

After both soft branches and all four s -cut boxes are summed, the result is

$$\langle \delta h^4 \rangle' \Big|_{s, \text{mixed}} = \frac{N_c g_t^4 H^4 k_s^3}{16 \prod_i k_i^3} \frac{2\mu(1 + \mu^2)}{3\pi \sinh(2\pi\mu)} \left[\mathcal{H}_{11}(-i\mu, +i\mu; 1) \right]^2. \quad (84)$$

It is independent of angles at this order. Equation (84) is the leading nonoscillatory nonanalytic term, whereas Eq. (76) is the leading mass-dependent clock signal. At $\mu = 0$ the separated branches coalesce; the limit must be taken after combining the relevant terms. Here $k_s^3 = (\mathbf{k}_s^2)^{3/2}$ is nonanalytic as a function of the soft momentum components, even though it has no mass-dependent oscillation. It cannot be discarded as a polynomial contact term.

For evaluation of the remaining mixed seed, the same radial identity and integration by parts give the general formula

$$\begin{aligned} \mathcal{H}_{11}(\alpha, \beta; 1) &= [\mathrm{i}(\alpha - 1) + \mu] \mathcal{H}_{21}(\alpha - 1, \beta; 1) \\ &\quad - \mathrm{i} \sum_{a,b=\pm} \left[\mathcal{S}_{ab}^{\alpha-1,\beta-2} + \mathrm{i}b \mathcal{S}_{ab}^{\alpha-1,\beta-1} \right]. \end{aligned} \quad (85)$$

Set $\alpha = -\mathrm{i}\mu + \delta$, $\beta = \mathrm{i}\mu + \delta$, perform the complete sum, and then take $\delta \rightarrow 0$ using the corrected regularized seeds. The conjugation relation $\mathcal{H}_{11}(\alpha, \beta; 1)^* = \mathcal{H}_{11}(\beta^*, \alpha^*; 1)$ proves the mixed seed is real. The code evaluates this limit by fourth-order extrapolation, with step-size and precision checks; the clock coefficients above instead use the analytic limit. For example, $\mathcal{H}_{11}(-0.73\mathrm{i}, +0.73\mathrm{i}; 1) = -1.388500280198814$; halving the extrapolation step changes this value by less than 2×10^{-15} .

Validation and limits. The new Whittaker implementation constructs all normal and anomalous contractions directly from Eq. (9). It verifies EOMs, normalization and the Hankel map at $\mu = 0.2, 0.73, 2, 5$. The independent eight-slot Wick enumerator gives 48 connected pairings and agrees with the cycle formula, and with half the Dirac result, for all 16 SK assignments at $\mu = 0.2, 0.73, 2$. Its test momenta are generic edge momenta: this is a local contraction test, not a momentum integral. The soft-expansion check retains the exact ${}_0F_1$ endpoints, Whittaker hard lines, both external energy shifts and all four orderings. On a finite time interval, at three masses and two geometries, its error relative to the leading nonzero coefficient decreases by approximately four when the soft momenta are halved. The nonzero single-cycle zeroth-order trace and its exchange cancellation are checked separately. Time-quadrature refinement is also tested.

A further independent calculation integrates the fermion Hankel kernels on BD-rotated contours at the physical mass indices: the two same-SK ordered triangles rotate together, and the two opposite-SK integrals rotate separately. It checks both diagonal and off-diagonal hard integrals and the hard-momentum derivative at $\mu = 0.2, 0.73$, with a common positive endpoint regulator $\delta = 0.8$, including a mixed-branch case. The derivative is obtained directly from the first-order mode equations, independently of the scalar bootstrap source; the regulated $\mathcal{P}_1, \mathcal{P}_2$ are compared as well. This tests the scalar-seed normalization beyond the conformal-scalar example. It does not directly evaluate the zero-regulator limit; that limit still uses the analytically summed seeds and regulator-path checks.

The retained scalar-seed tests check the elementary conformal-scalar normalization, Gamma reductions, regulator paths and working precision. Comparing the regulated and analytically summed versions of the same seed is a consistency check, not a second derivation of that seed. Neither finite-interval soft tests nor special-function agreement is a direct numerical integration of the full unexpanded BD box. The analytic contact part and its renormalization, overlapping collapsed channels, evolving masses, nonzero chemical potential and curvature conversion have not been computed. Reproducible commands and residuals are recorded in `checks/top_quark_audit.md`.

A Dirac comparison as a check

This appendix introduces four-component notation only to check the calculation above. Define

$$\eta = \begin{pmatrix} \psi_\alpha \\ \psi^{\dagger\dot{\alpha}} \end{pmatrix}, \quad \zeta = \begin{pmatrix} \psi_\alpha \\ \psi^{\dagger\dot{\alpha}} \end{pmatrix} = T\eta, \quad T = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & E \end{pmatrix}, \quad X = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix}. \quad (86)$$

If $C = \langle T_C \zeta \zeta^T \rangle$, then $R_W = \langle T_C \eta \eta^\dagger \rangle = T^\dagger C X T$ and $T^\dagger X J T = \beta_W = X$. Cyclicity of the trace proves, without any momentum approximation,

$$\text{tr}_4[JC_{ij}JC_{jk}JC_{kl}JC_{li}] = \text{tr}_4[\beta_W R_{W,ij}\beta_W R_{W,jk}\beta_W R_{W,kl}\beta_W R_{W,li}]. \quad (87)$$

The left-hand side is exactly the 16 two-component type contractions in Eq. (20). With

$$U_B = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{1}_2 & -\mathbf{1}_2 \\ \mathbf{1}_2 & \mathbf{1}_2 \end{pmatrix}, \quad R = U_B^\dagger R_W U_B, \quad \beta = U_B^\dagger \beta_W U_B = \text{diag}(\mathbf{1}_2, -\mathbf{1}_2), \quad (88)$$

its entries are precisely the projections of Eq. (24). The positive and independent negative frequency mode matrices in this basis may be chosen as

$$U = \begin{pmatrix} F\mathbf{1}_2 \\ G\sigma_k \end{pmatrix}, \quad V = \begin{pmatrix} G^*\mathbf{1}_2 \\ -F^*\sigma_k \end{pmatrix}, \quad R^> = U_1 U_2^\dagger, \quad R^< = -V_1 V_2^\dagger. \quad (89)$$

The negative-frequency V is unrelated to the name of Aoki's radial v_λ ; its relative minus sign is essential. For the charged Dirac field the particle and antiparticle operators are independent, whereas in η they are related by charge conjugation. The two-point matrix is nevertheless the same. The distinction in a closed loop is the Majorana 1/2 established by Wick counting. Multiplying by two copies recovers six Dirac cycles with coefficient $-N_c g_t^4$ each. There is no remaining orientation, spin or color multiplicity to add.

References

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